

H1 2011 Preliminary Examination Paper 1 (Suggested Solutions)

Paper 1

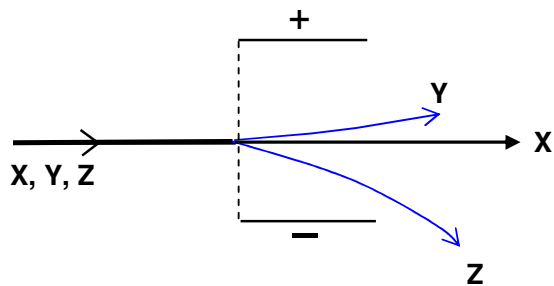
1	B	2	B	3	A	4	A	5	C
6	D	7	C	8	C	9	A	10	D
11	C	12	A	13	C	14	B	15	C
16	B	17	B	18	A	19	B	20	D
21	D	22	D	23	D	24	A	25	A
26	B	27	C	28	B	29	C	30	D

Paper 2 - Section A

1(a)

Particle	Electric charge	Mass number	Number of		
			Protons	Electrons	Neutrons
X	0	32	16	16	16
Y	1-	81	35	36	46
Z	3+	70	31	28	39

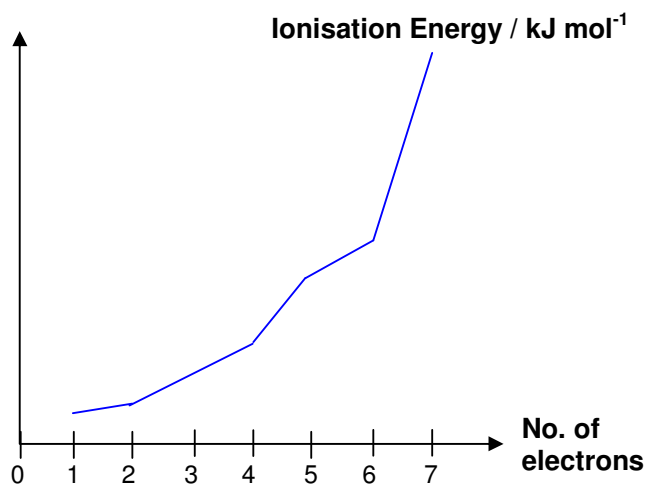
(b)



(c)(i) X: $1s^2 2s^2 2p^6 3s^2 3p^4$

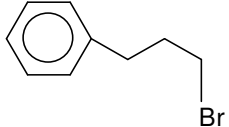
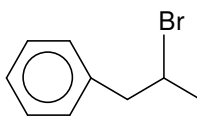
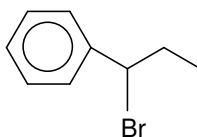
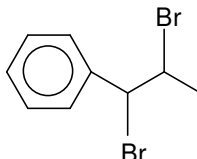
(ii) Z: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$

(d)

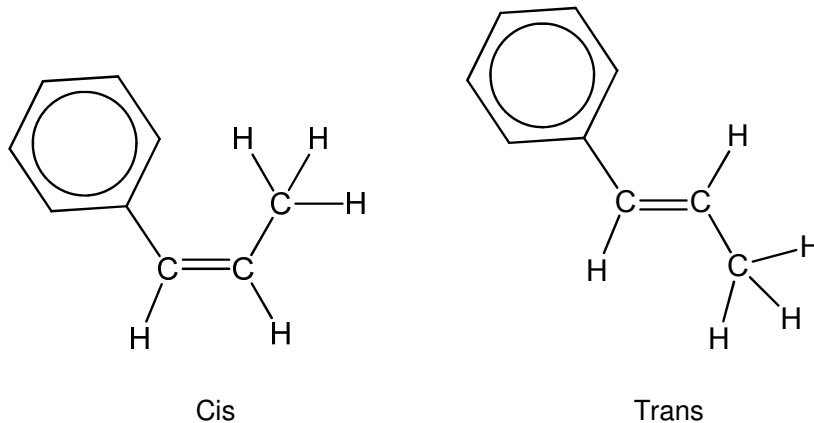


2(a)(i) Ethanolic KOH, heat

(ii) Br_2 in CCl_4

A 	D 
B 	E & F 
C 	G 

(b) E & F exhibit geometrical isomerism.



3(a)(i) The **bond energy** of a X–Y bond is the **average energy** absorbed when **1 mole of X–Y bonds are broken in gaseous state**.

(a)(ii) $\text{BE (C–Cl)} = 340 \text{ kJ mol}^{-1}$
 $\text{BE (O–H)} = 460 \text{ kJ mol}^{-1}$
 $\text{BE (C–O)} = 360 \text{ kJ mol}^{-1}$
 $\text{BE (H–Cl)} = 431 \text{ kJ mol}^{-1}$

$$\begin{aligned}\Delta H_r &= \Sigma \text{BE of bonds broken in reactants} - \Sigma \text{BE of bonds formed in products} \\ &= 2(340) + 2(460) - 2(360) - 2(431) \\ &= \mathbf{+18 \text{ kJ mol}^{-1}} \text{ (3sf)}\end{aligned}$$

- (b) This is because DEHP is **more soluble in non-polar solvents** than polar solvents.

Due to the presence of large hydrophobic groups (benzene ring and long alkyl chain), DEHP is able to form favourable **instantaneous dipole-induced dipole (id-id)** interactions with non-polar solvents. Hence, it is readily soluble in non-polar solvents.

However, in polar solvents like water, the **id-id** interactions formed between DEHP and water molecules are **weaker** than the **hydrogen bonds** formed between water molecules. Hence, DEHP is less soluble in polar solvents like water.

- (c)(i) Reagent: dilute $\text{H}_2\text{SO}_4/\text{HCl}/\text{HNO}_3$
Condition: Heat

- (c)(ii) Test:
Add acidified $\text{KMnO}_4(\text{aq})$ [or acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$] to each compound and heat.

Observations:

For 2-ethylhexanol, purple KMnO_4 solution is decolourised (or orange $\text{K}_2\text{Cr}_2\text{O}_7$ solution turns green).

For phthalic acid, KMnO_4 remains purple (or orange for $\text{K}_2\text{Cr}_2\text{O}_7$).

Alternative test:

Add $\text{Na}_2\text{CO}_3(\text{aq})$ to each compound.

Observations:

For phthalic acid, effervescence is observed and colourless gas evolved gives a white ppt with $\text{Ca}(\text{OH})_2(\text{aq})$.

For 2-ethylhexanol, no effervescence is observed and no gas is evolved.

- (d)(i) Molar mass of 2-ethylhexanol = $8(12.0) + 18(1.0) + 16.0$
= 130 g mol^{-1}

Amount of 2-ethylhexanol = $1.5 / 130$
= $0.01154 \text{ mol (4sf)}$

$$\begin{aligned}\Delta H_c^\ominus &= -57 / 0.01154 \\ &= -4939 \text{ kJ mol}^{-1} \text{ (4sf)} \\ &= \mathbf{-4940 \text{ kJ mol}^{-1} \text{ (3sf)}}\end{aligned}$$

- (d)(ii) By Hess' law,
 $\Delta H_f^\ominus (\text{C}_8\text{H}_{18}\text{O}) + \Delta H_c^\ominus (\text{C}_8\text{H}_{18}\text{O}) = [8 \times \Delta H_c^\ominus (\text{C})] + [9 \times \Delta H_f^\ominus (\text{H}_2\text{O})]$
 $\Delta H_f^\ominus (\text{C}_8\text{H}_{18}\text{O}) - 4939 = 8(-393.5) + 9(-285.9)$
 $\Delta H_f^\ominus (\text{C}_8\text{H}_{18}\text{O}) = \mathbf{-782 \text{ kJ mol}^{-1} \text{ (3sf)}}$

- 4(a)(i) $[\text{H}^+(\text{aq})]$ from $0.100 \text{ mol dm}^{-3} \text{ H}_2\text{CO}_3 = 10^{-3.68} = 2.09 \times 10^{-4} \text{ mol dm}^{-3}$

Since the concentration of $\text{H}^+(\text{aq})$ is less than the concentration of H_2CO_3 , H_2CO_3 undergoes partial dissociation. Hence, H_2CO_3 is a weak acid.

- (a)(ii) Increase in $[\text{H}^+(\text{aq})] = 10^{-8.14} - 10^{-8.25}$

$$= (7.24 \times 10^{-9}) - (5.62 \times 10^{-9}) \text{ mol dm}^{-3}$$

$$= 1.62 \times 10^{-9} \text{ mol dm}^{-3}$$

$$\text{Percentage increase in } [\text{H}^+(\text{aq})] = \frac{1.62 \times 10^{-9}}{5.62 \times 10^{-9}} \times 100\% = 28.8\%$$

- (a)(iii)** Upon addition of small amount of H^+ ,
 $\text{H}^+ + \text{CO}_3^{2-} \rightarrow \text{HCO}_3^-$

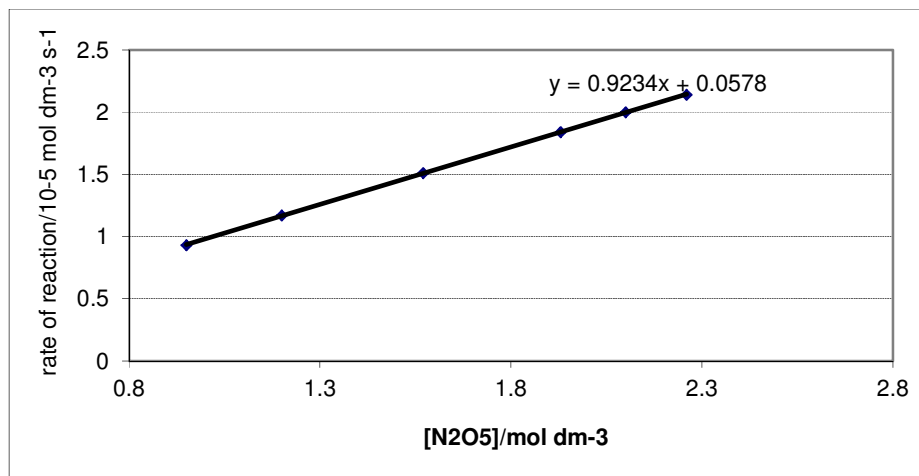
Upon addition of small amount of OH^- ,
 $\text{OH}^- + \text{HCO}_3^- \rightarrow \text{H}_2\text{O} + \text{CO}_3^{2-}$

- (a)(iv)** Acid rain arising from dissolved oxides of nitrogen and SO_2 in the atmosphere.
- (b)(i)** If a system in equilibrium is subjected to a change (or stress) which disturbs the equilibrium, the system responds in such a way as to counteract the effect of the change (or stress).
- (b)(ii)** When concentration of CO_2 increases, the equilibrium position in $\text{CO}_2 + \text{CO}_3^{2-} + \text{H}_2\text{O} \rightleftharpoons 2 \text{HCO}_3^-$ shifts to the right to decrease the concentration of CO_2 .

This causes the concentration of CO_3^{2-} to decrease. Hence, the equilibrium position in $\text{Ca}^{2+} + \text{CO}_3^{2-} \rightleftharpoons \text{CaCO}_3$ to shift to the left. This, in turn, causes a decrease in amount of CaCO_3 that can be used for building (or calcification) of shells and coral reefs.

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5(a)(i)



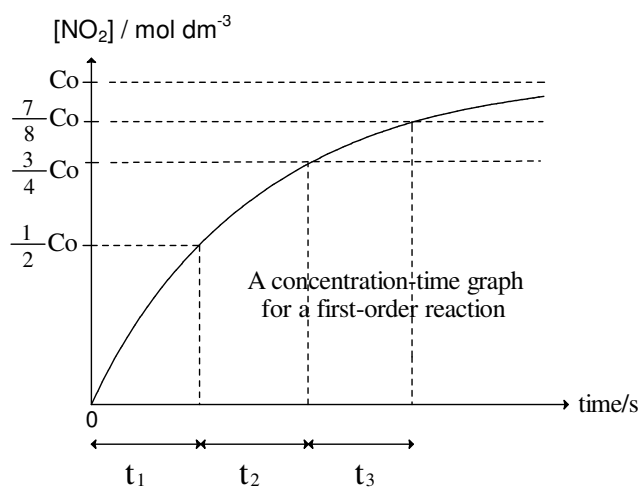
(a)(ii) Order of reaction wrt N₂O₅ is 1.

The rate of reaction is directly proportional to concentration of N₂O₅.

(a)(iii) Rate = $k[\text{N}_2\text{O}_5]$

(a)(iv) Rate constant = gradient of graph = $9.23 \times 10^{-6} \text{ s}^{-1}$

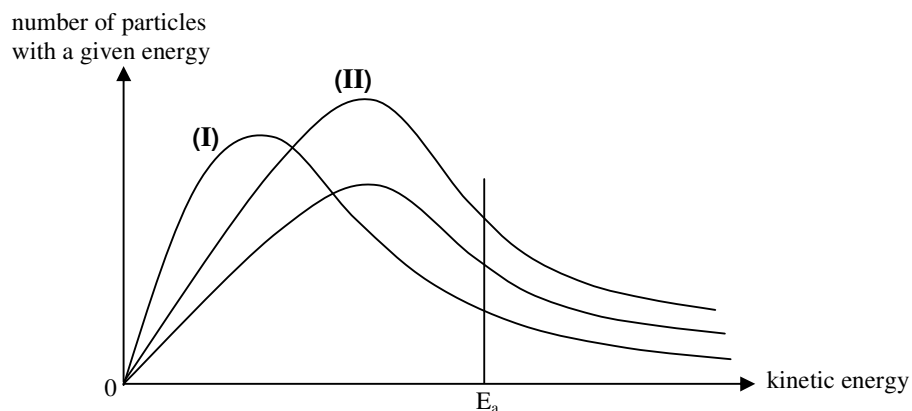
(a)(v)



where $t_1 = t_2 = t_3$

Constant half life must be indicated correctly.

(b)(i)



(b)(ii) As the concentration of a reactant increases, the reactant particles are generally closer together.

Thus, the frequency of collisions between reactant particles increases.

From the graph in (ii), the number or proportion of reactant particles possessing energy greater than or equal to the activation energy also increases. Hence, the frequency of effective collisions increases and the reaction rate increases.

(c)(i) If the total pressure of the equilibrium mixture is increased, the system will try to counteract the increase in total pressure by favouring reaction that produces fewer molecules of gases.

The position of equilibrium in $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ shifts to the right causing the percentage of NH_3 to increase with increasing pressure.

(c)(ii) The production of NH_3 is an exothermic process since at higher temperatures, there is a lower percentage of NH_3 at equilibrium.

Hence the backward reaction is endothermic since the endothermic reaction is favoured at higher temperatures i.e. forward reaction producing NH_3 is exothermic.

(c)(iii) The percentage of NH_3 at equilibrium remains unchanged.

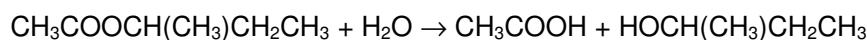
A catalyst lowers the activation energy of both the forward and backward reactions to the same extent.

OR

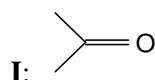
A catalyst increases the rate of the forward and backward reactions to the same extent:

6(a)(i) $\text{C}_6\text{H}_{12}\text{O}_2$

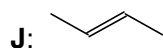
- (ii) Dil. HCl, heat
- (iii) M_r of **C** = 60.6
C is ethanoic acid.
- (iv) **D** contains the $\text{CH}_3\text{CH}(\text{OH})-$ group, and **F** contains $\text{CH}_3\text{CO}-$ group, hence both react with alkaline aq. iodine to form CHI_3 and carboxylate salts. **E** is CHI_3 .
- (v) **D** is $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{OH}$.
F is $\text{CH}_3\text{CH}_2\text{COCH}_3$.
G is $\text{CH}_3\text{CH}_2\text{COO}^- \text{Na}^+$
- (vi) **B** is $\text{CH}_3\text{COOCH}(\text{CH}_3)\text{CH}_2\text{CH}_3$.



- (b)(i) Step 1: KMnO_4 , dil. H_2SO_4 , heat
Step 2: LiAlH_4 in dry ether (or H_2 over Ni catalyst at high temp & pressure)



- (ii) Step 1: ethanolic KOH, heat
Step 2: cold dil. KMnO_4 , aq. NaOH



- (c) 2,3-butanediol has a much higher boiling point than 2-chlorobutane despite their M_r values being close.
Both exist as simple covalent molecules with weak intermolecular interactions, but the stronger hydrogen bonds between 2,3-butanediol molecules overcome during boiling require more energy than the weaker permanent dipole-permanent dipole forces overcome between 2-chlorobutane molecules.

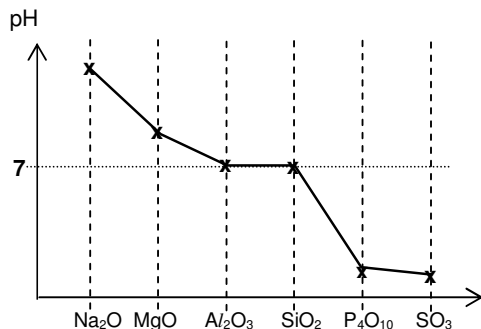
- 7(a)(i) Both Na_2O and MgO have **giant ionic structures** with strong electrostatic forces of attractions between the oppositely charged ions.

However, due to the **higher charge** and **smaller ionic radius** of Mg^{2+} as compared to Na^+ , MgO has **stronger ionic bonds** than Na_2O . **More energy** is required to break the ionic bonds in MgO than Na_2O . Hence, MgO has a higher melting point than Na_2O .

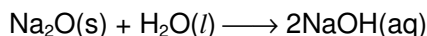
- (a)(ii) SiO_2 has a **giant covalent structure** with **strong covalent Si–O bonds** whereas SO_3 has a **simple molecular structure** with **weak intermolecular van der Waals' forces of attraction/ instantaneous dipole–induced dipole (id–id) interactions**.

Hence, **larger amount of energy** is required to break the strong covalent bonds in SiO_2 than the weak id–id interactions in SO_3 . Hence, SiO_2 has a much higher melting point than SO_3 .

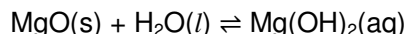
(a)(iii)



(a)(iv) Na₂O is an **ionic oxide** and will react vigorously with water to form **strongly alkaline solutions**.

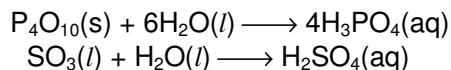


MgO is also an **ionic oxide** but reacts less readily with water as MgO is only slightly soluble in water due to its highly exothermic lattice energy. Thus, the solution formed is **weakly alkaline**.



Al₂O₃ does not dissolve in water because its lattice energy is highly exothermic. Likewise, **SiO₂** does not dissolve in water because of the strong Si–O covalent bonds in the giant covalent structure. Since these oxides do not dissolve in water, **pH will remain neutral at 7** (at 25 °C).

P₄O₁₀ and SO₃ are **non-metal oxides**. Hence, they will react readily with water to form **strongly acidic** solutions.



(b)(i) M_r of AgCl = 108 + 35.5 = 143.5

Amount of AgCl formed initially
= 1.55 / 143.5
= 0.01080 mol (4sf)

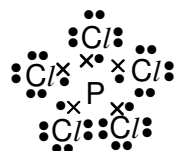
Amount of AgCl formed after **RCI_x** is heated
= 0.93 / 143.5
= 0.006481 mol (4sf)

Mole ratio of AgCl formed in Experiment 1 & 2:
= 0.01080 : 0.006481
= 1.67 : 1
= **5 : 3** → This suggests that $x = 5$.

On heating, **RCI₅** loses Cl₂ to form **RCI₃**.
Thus, **R** belongs to **Group V**.

- (b)(ii) Mole ratio of $\text{RCI}_5 : \text{AgCl} = 1 : 5$
Amount of $\text{RCI}_5 = 0.01080 / 5$
 $= 0.002160 \text{ mol (4sf)}$
 M_r of $\text{RCI}_5 = 0.45 / 0.002160$
 $= 208.3 \text{ (4sf)}$
 A_r of $\text{R} = 208.3 - 5(35.5)$
 $= 30.8 \approx 31 \rightarrow \text{R is phosphorus}$
Hence, RCI_x is PCl_5 .

(b)(iii)



Molecular geometry: Trigonal bipyramidal

- (c)(i) $5\text{CO} + \text{I}_2\text{O}_5 \longrightarrow \text{I}_2 + 5\text{CO}_2$
Type of reaction: Redox
- (c)(ii) Amount of $\text{S}_2\text{O}_3^{2-}$ required $= 15.0/1000 \times 0.10 = 0.0015 \text{ mol}$
Mole ratio of $\text{S}_2\text{O}_3^{2-} : \text{I}_2 : \text{CO} = 2 : 1 : 5$
Amount of CO present $= 0.0015 / 2 \times 5 = 0.00375 \text{ mol}$
 M_r of $\text{CO} = 12.0 + 16.0 = 28.0$
Concentration of $\text{CO} = (0.00375 \times 28) / 100 \times 1000 = 1.05 \text{ g dm}^{-3} \text{ (3sf)}$